

Thursday 2026-05-28 PM (Session 2) Parallel Work Menu v0.1

Keeper (per Casey directive — afternoon session restart)

2026-05-28 Thursday 16:14 EDT

Thursday PM Parallel Work Menu (Session 2)

How to run this

1. Pull continuously — bank a batch of results, then report a batch. Not one-and-stop.
2. Do NOT wait on each other — every lane has independent work. The only cross-CI gate is the A1 Macdonald parameter-role (below), and even that doesn't block most items.
3. Show all threads — the tier-honest brake controls the *claim*, not the *investigation*. Keep leads live (quark mass ratios, fermion winding-web), pursue their mechanisms, weave the story after.
4. Log NEW investigation areas + WHY in RUNNING_NOTES as you go. Casey harvests them for the next list.
5. Tier discipline stands (scheme-invariance vs coincidence-denominator axes; no premature RATIFIED). Standing conventions: three-genus (Hua5/FK6/embed7), α -disambiguation (Jack 2/3, fine 1/137, Macdonald t), and the new Macdonald-parameter convention once Lyra fixes it.

THE ONE GATE (clears first — unblocks A1, the Casey-PRIMARY paper)

Macdonald parameter-role reconciliation (Keeper_Macdonald_Parameter_Role_Flag_v0_1.md): the substrate base 2 is the Macdonald t (Hall-Littlewood corner, $q=0$), NOT the Macdonald q; A1's "($q=2$, $t=1/137$)" is HELD. Lyra leads, Elie verifies numerically, Keeper grades. Until it clears, A1's central parameterization stays held — but every other A1 section (Serre constants, mixing-angle spine, confinement, three-genus) is forward and can advance.

LYRA — theory + papers (~15 items; A1-gate first, then forward)

1. A1 Macdonald parameter-role fix (the gate): is the Hall algebra Hall-Littlewood ($q=0$, $t=2$)? Where does $\alpha=1/137$ actually enter (Jack corner / affine level / $N_{\max}$)? Correct A1's parameterization; draft the Macdonald-parameter convention.
2. Koornwinder single-object theorem (with Elie): one explicit object with proved limits to the FK/Heckman-Opdam spherical functions (Jack corner) AND the Ringel-Hall algebra (Hall-Littlewood corner). The unification made into a single proved object.
3. A1 v0.3 — re-derive the Hall structure constants at the corrected corner; fold in ρ -vector + c_{FK} -derived-theorem + three-genus.
4. Cyclotomic $\leftrightarrow$ Coxeter forcing — turn generation count matched $\rightarrow$ forced (commitment-cycle period = $h(B_2)=4$). The deepest open gate; B1's promotion target.
5. Strong-Uniqueness v1.1 — Cal #143's 4 flags, now strengthened by c_{FK} -derived + ρ -vector pins-3-primaries + Route-A.
6. Substrate-privileged-scheme hypothesis — the promotion path for the quark mass-ratio leads (is there a substrate-natural scale where they're exact?). Keep the thread live per show-all-threads.
7. B-series drafts: B2 (TMAP/OMAP), B3 (quark spectrum), B4 (lepton), B5 (charge), B6 (bosons+Higgs+mixing).
8. A2 (Quiver Q_{B2}) + A4 (Holographic operators) drafts.

9. C1 (Λ) + C2 (spin-statistics) — advance the existing v0.1 drafts.
10. Fermion winding-web (Casey seed) — formalize: do the scheme-invariant ratios compose into a connected web with few winding-step generators?
11. WCGP v0.2 + Vol 16 A_sub chapters.
12. Any remaining 14-paper drafts as appetite allows.

ELIE — toys (next toy 3588; ~12 items)

1. Macdonald parameter-role verification (feeds the gate): compute structure constants at Hall-Littlewood ($q=0, t=2$) vs ($q=2, t=1/137$); which reproduces the substrate Serre + observable content? Decisive for A1.
2. Koornwinder single-object numerical (with Lyra): evaluate one object's limits to both corners on shared partitions.
3. Generation-forcing numerical test — commitment-cycle as Coxeter element of order $h(B_2)=4$; does it force chain termination?
4. Substrate-privileged-scheme scan — sweep scales; is there one where quark ratios hit substrate-natural values exactly?
5. Jack-bridge extension — higher-degree partitions, more limits.
6. ρ -vector consequences — what other spectral quantities does $\rho = (n_C, N_C)/\text{rank pin}$?
7. Quark sector battery at the corrected Macdonald corner.
8. K-type graph Phase C+ + visualization.
9. K52a Session 8+ (BCS / substrate-Hamiltonian).
10. Verification harnesses for Lyra's A1 v0.3 + Koornwinder + Strong-Uniqueness v1.1.
11. GF(32) deep cyclotomic.
12. Bosons-as-coupling numerical checks.

GRACE — catalog (next INV ~5269; ~10 items)

1. Macdonald-parameter convention catalog entry (once Lyra resolves) — tag which substrate quantity is q , which is t , which corner. Fourth standing convention.
2. Phase 2 SPLP weekly batch.
3. Catalog-wide validating-axis labeling (the defensibility pass Casey approved as option (a)) — name scheme-invariance vs coincidence-denominator per forward claim.
4. Coincidence-denominator metric catalog-wide (Grace's nominated sweep) — systematic forward-vs-fitted triage.
5. Bergman anchors at corrected $5/2$ — extend.
6. ρ -vector / two-region tagging extension.
7. Quark thread-map maintenance (scheme behavior, kept live).
8. Weinberg-family scan (other $\text{rank}+g=N_C^2$ -type partitions).
9. 9-element operational set full tagging.
10. Cal #19 compliance on external-facing counts.

CAL — referee (own cadence; ~6 items)

1. PMNS form-typing ($/N_{\max}$ vs T1935) — close it; it's the last open item from the morning.
2. Type the Macdonald parameter-role resolution when Lyra delivers (A/B/C corner question).
3. Cold-read A1 v0.3 once the parameter-role clears.
4. Cold-read the Koornwinder / two-corner unification.
5. Strong-Uniqueness v1.1 review.
6. Cal #32 candidate watch: is “parameter-role / which-corner” a new calibration class worth standing (like three-genus)?

KEEPER — me (audit + coordination)

1. Grade the Macdonald parameter-role resolution (gates A1 — my audit when Lyra + Elie deliver).

2. Grade the Koornwinder single-object theorem when it lands.
 3. K-audit pre-stages on the day's FRAMEWORK-PLUS results (ρ -vector, two-corner, c_{FK} -derived-theorem, Coxeter generations).
 4. Confirm K67/T2401 + T2440 relabel-only (with Grace's sweep).
 5. Investigation Board v0.2.
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NEW AREAS log — append as you work (area + WHY, one line each)

- (Keeper seed) Macdonald-parameter convention as the 4th standing guard — WHY: the $q \leftrightarrow t$ role-confusion just gated the PRIMARY paper; a one-line convention fences the whole class, like three-genus did.
- (Keeper seed) Does the two-corner picture predict a *third* corner observable? — WHY: if geometry=Jack and Hall algebra=Hall-Littlewood are two corners of the Macdonald square, the other two corners (Schur $q=t$, Macdonald-generic interior) may carry substrate meaning worth identifying.

— Keeper, 2026-05-28 Thursday PM. Pull continuously; show all threads; clear the A1 gate first; log new areas.